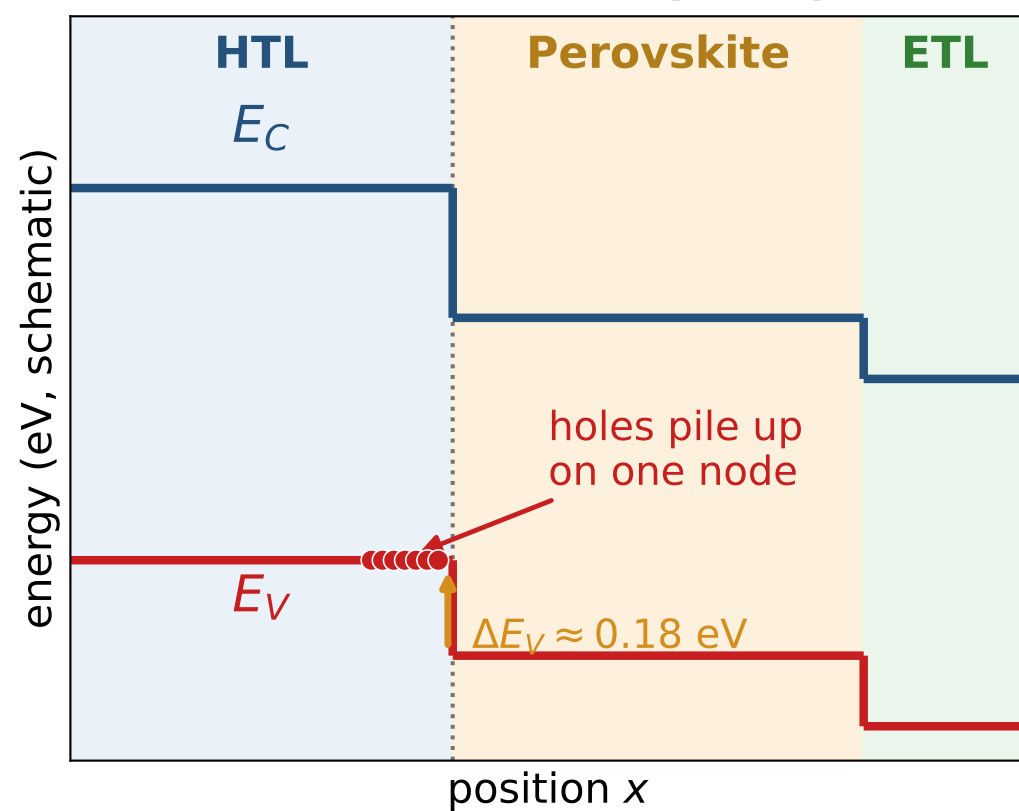


Heterointerface Auger de-spike f — principle and matrix response

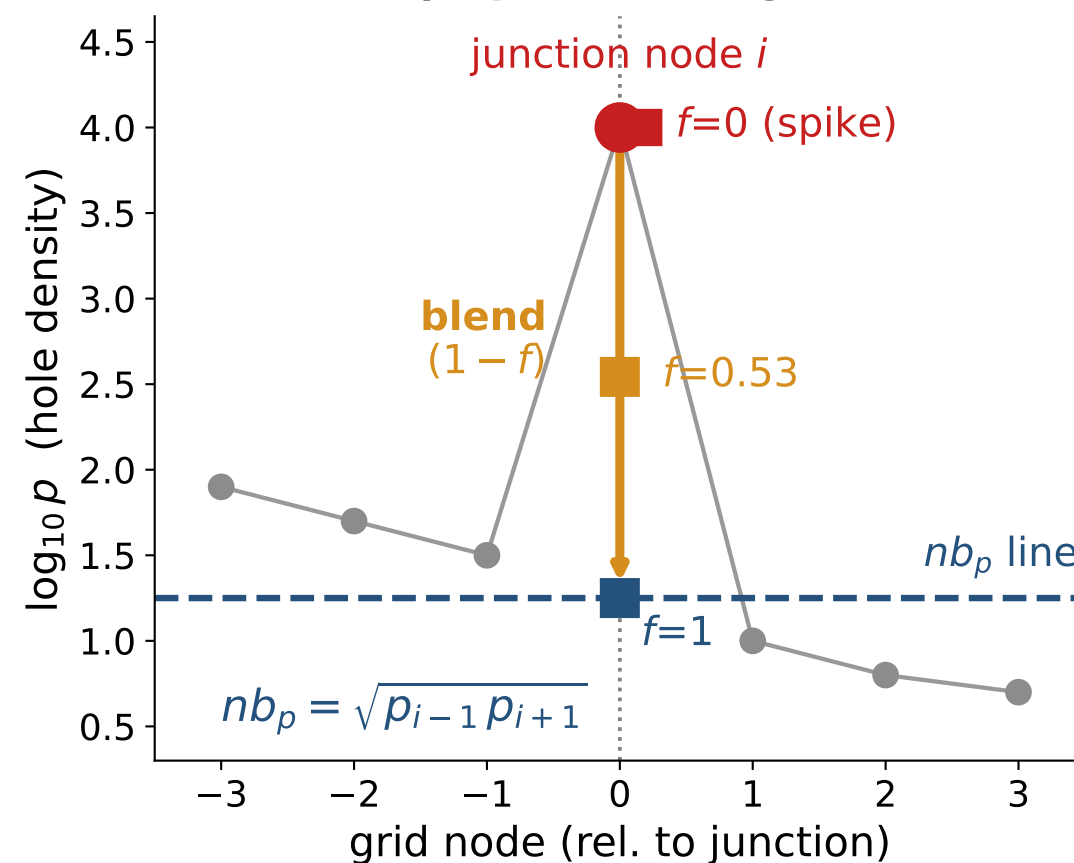
PRINCIPLE (top): why a spike forms and what f removes.

MATRIX TEST (bottom): how the base operating point and the 10 SCAPS sweeps respond to f .

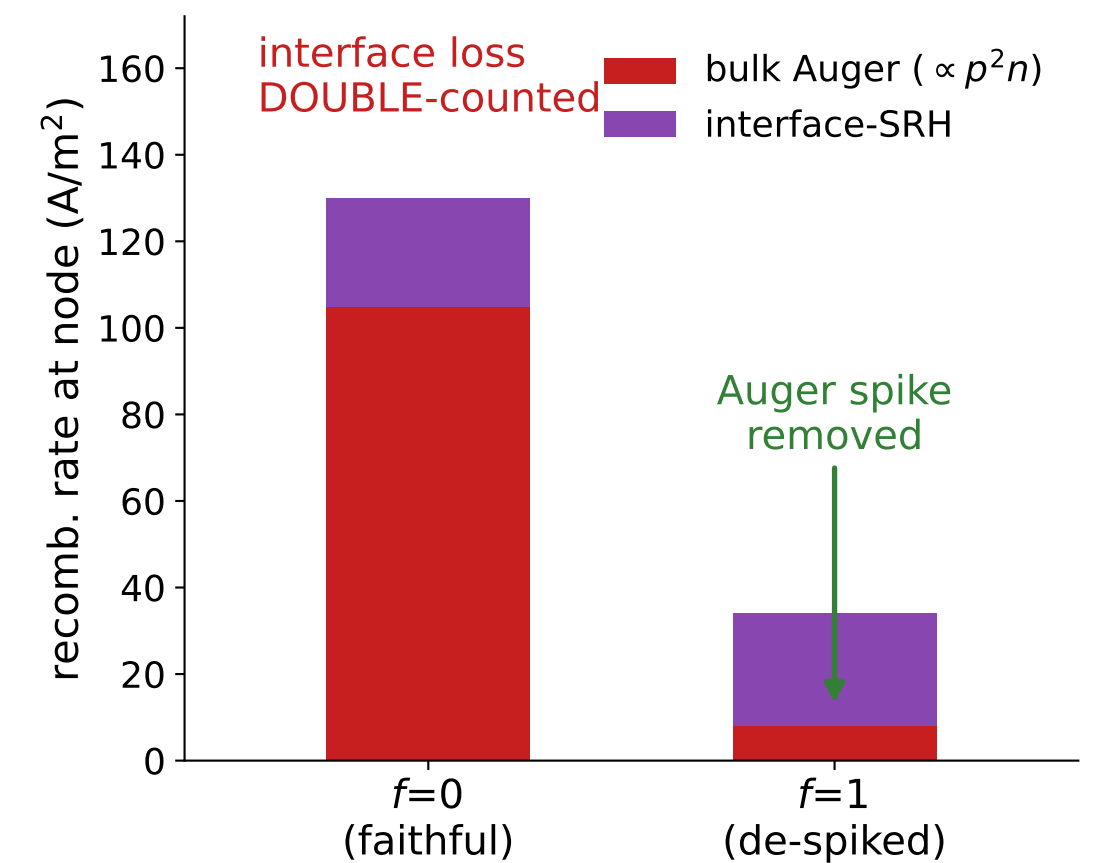
(A) Band offset → hole pile-up



(B) Density spike → neighbour-mean blend



(C) Recombination at the node



$$p_{rec}[i] = nb_p + (p[i] - nb_p)(1 - f)$$

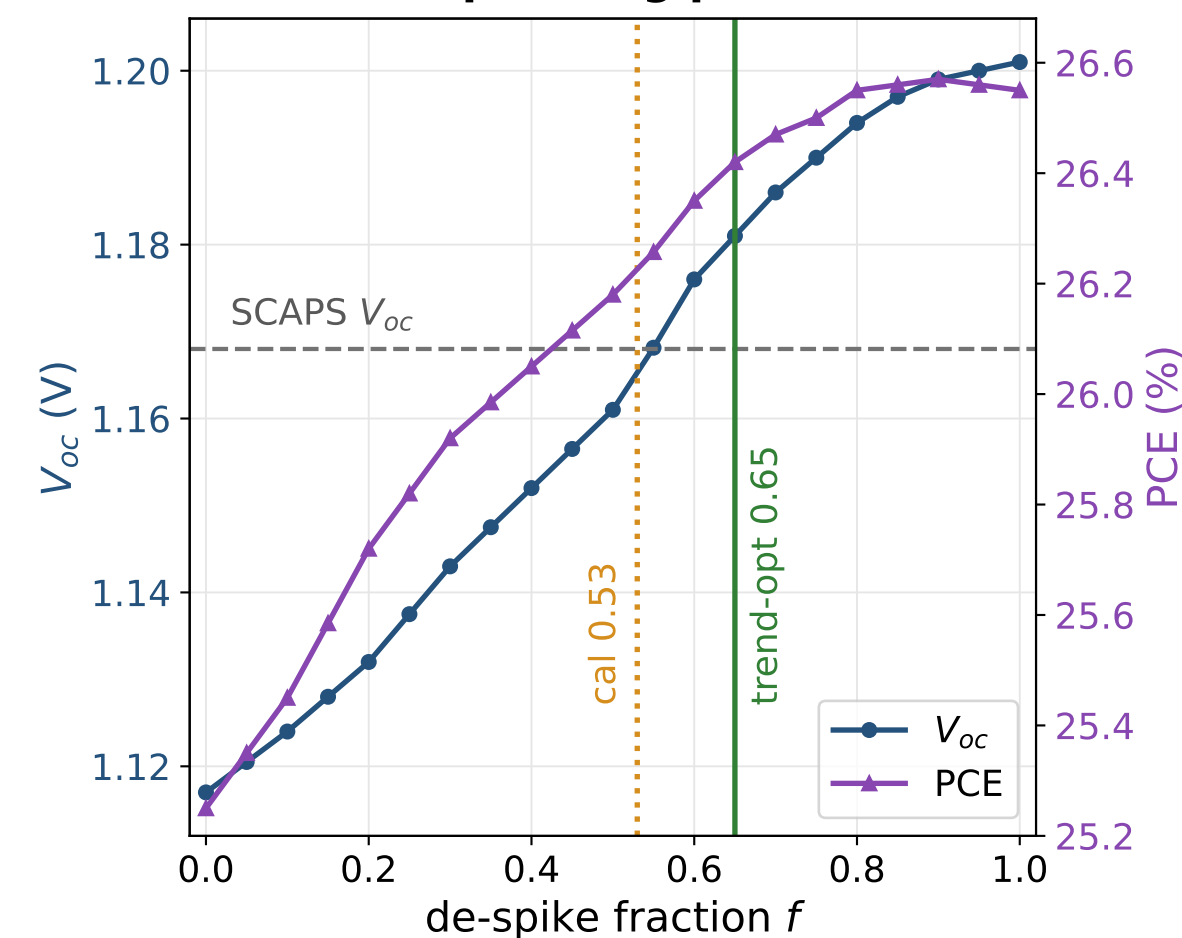
$$R = SRH + rad + Auger(n_{rec}, p_{rec})$$

$$C_n, C_p \neq 0$$

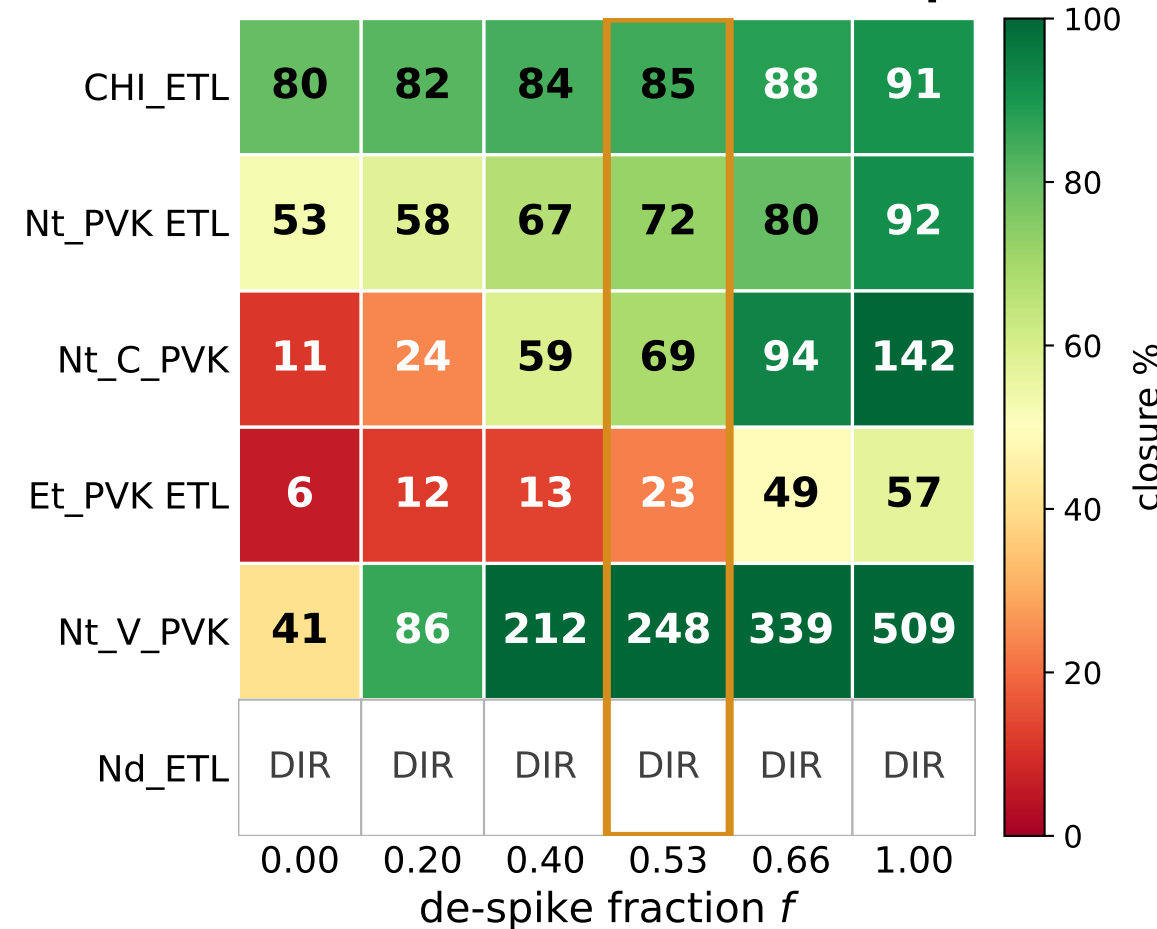
Only the recombination-rate density at the junction node is blended; transport / SG flux uses the true (spiked) density $\Rightarrow$ current stays physical.

$f=0$ over-counts · $f=1$ removes the pile-up · value is TUNED to SCAPS (emulation knob, default OFF).

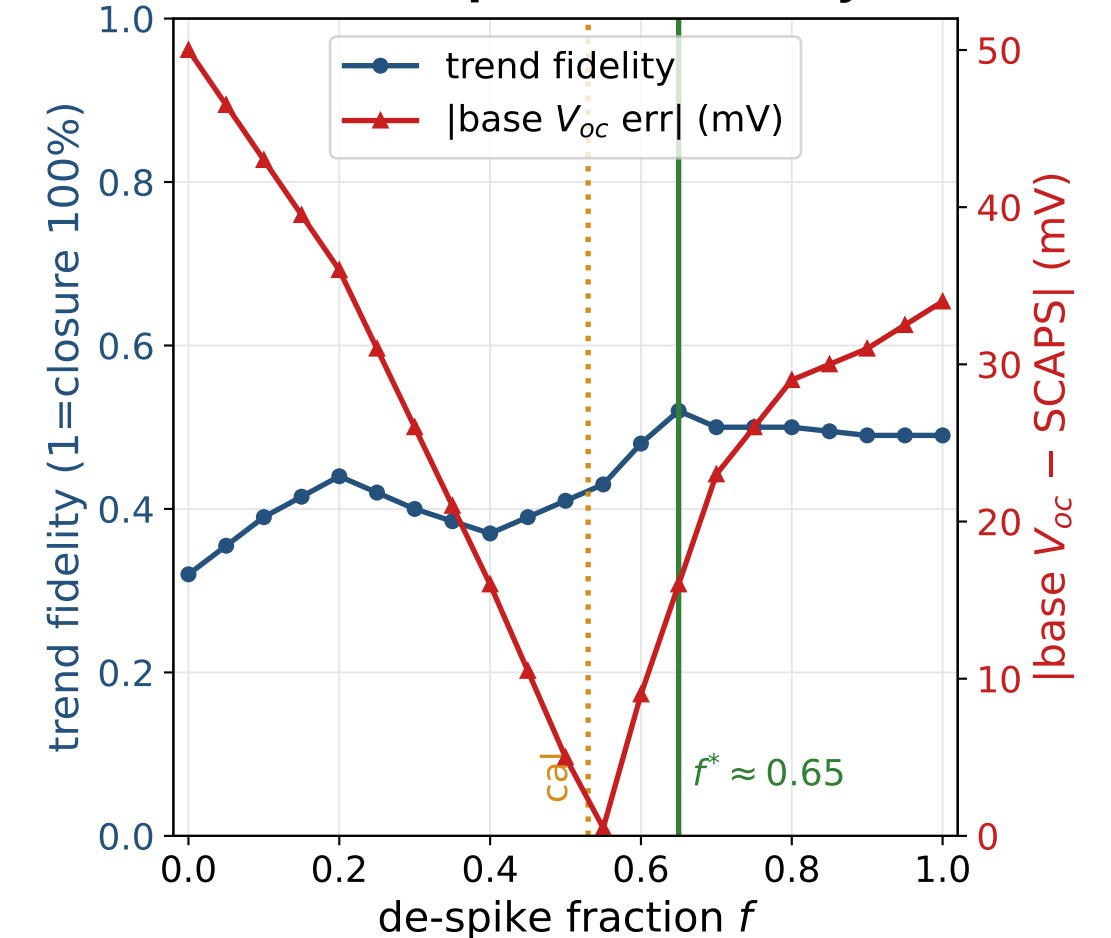
(D) Base operating point vs f



(E) Trend closure% — sweep $\times f$



(F) Trend-optimal f : fidelity vs base cost



Solar-cell reading: the over-counted junction Auger artificially depresses V_{oc} AND compresses every sweep's V_{oc} range.

Removing it ($\uparrow f$) raises base V_{oc} past SCAPS near $f \approx 0.55$ and OPENS the sweep ranges (closure $\uparrow$). Nd_ETL & HTL/PVK stay direction-locked. Trend-optimum $f^* \approx 0.65$.